AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.



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PART 1: Multiple Choice Questions [10 Marks]

Questions 1 [10 marks] (1 mark/each):

1. Which of the following best describes the primary objective of information security?
   A. To maximize system performance
   B. To protect confidentiality, integrity, and availability
   C. To improve user interface design
   D. To reduce the amount of stored data

2. A passive attack mainly involves:
   A. Modifying stored data
   B. Disrupting system services
   C. Eavesdropping or monitoring communications
   D. Changing user permissions

3. Asymmetric encryption uses:
   A. One shared secret key
   B. Two related keys
   C. No keys
   D. Only hash values

4. A cryptographic hash function produces:
   A. A fixed-length output from arbitrary input
   B. A public and private key pair
   C. A decrypted version of ciphertext
   D. An access control list

5. Privacy by Design requires privacy requirements to be considered:
   A. Only after a complaint is received
   B. Only during final testing
   C. Throughout the system development process
   D. Only during maintenance

6. Personally Identifiable Information (PII) is information that can:
   A. Only describe software behavior
   B. Distinguish or trace an individual's identity
   C. Only identify a company
   D. Never be sensitive

7. Which of the following is an example of sensitive PII?
   A. Job title
   B. Public department name
   C. Medical information
   D. Generic website name

8. GDPR stands for:
   A. General Data Processing Regulation
   B. General Data Protection Regulation
   C. Global Data Privacy Rules
   D. Government Data Privacy Regulation

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9. The National Vulnerability Database (NVD) mainly contains:
   A. Known technical vulnerabilities
   B. User passwords
   C. Private medical records
   D. Software license prices

10. Role-Based Access Control (RBAC) assigns permissions based on:
   A. User roles
   B. Password length only
   C. Random access requests
   D. Storage capacity

PART 2: Short Answer Questions [20 Marks]

P2-Q1 [5 marks]: What is the difference between passive attacks and active attacks, giving one example of each.

P2-Q2 [5 marks]: What is Privacy by Design? and how it differs from privacy engineering.

P2-Q3 [5 marks]: Compare de-identification and anonymization of data, including their purposes and limitations.

P2-Q4 [5 marks]: Explain the principle of least privilege.

PART 3: Problem Solving and Case Studies [30 Marks]

P3-Q1 [20 marks]: Study the following cases and answer the related questions:

CASE-1 [4 marks]: **"A school management system stores students' grades electronically. A staff member's password is stolen, and the attacker logs in and changes several students' grades"**. Suggest two security mechanisms to reduce the chance of this attack happening again.

CASE-2 [4 marks]: **"A new supplier and a company need to exchange confidential contract documents online, but they have never shared a secret key before".** Which type of encryption should be used initially? Explain why

CASE-3 [4 marks]: **"A mobile learning application collects students' names, locations, and usage behavior. The developers only add privacy settings after parents complain about excessive data collection".** What privacy principle is violated?

CASE-4 [4 marks]: **"A smart parking application collects drivers' location data to help them find available parking spaces. Later, the company shares this location data with marketing companies without clearly informing users".** Identify the privacy threat and suggest a privacy-respecting alternative.

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CASE-5 [4 marks]: **"An online university portal has many cases where students are tricked into giving away their passwords through fake login pages".** Recommend a stronger authentication method and explain how it improves security.

P3-Q2 [10 marks]: A company document management system controls access to different business resources. The system has the following subjects and objects. Construct the Access Control Matrix for this system.

Subjects (Users)
- HR Manager
- Project Manager
- Employee

Objects (Resources)
- Employee Records
- Project Documents
- System Reports

Access Rights
- Read (R)
- Write (W)
- Delete (D)

The security policy specifies:
- The HR Manager has full access to Employee Records and read-only access to System Reports.
- The Project Manager has Read/Write access to Project Documents and read-only access to System Reports.
- The Employee has read-only access to Project Documents.
- Only the HR Manager can access Employee Records.
- No user except the HR Manager can delete Employee Records

End of questions



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